

AV_Culpability_E13_ForceDynamics - POSTED

Start of Block: informed_consent

ic Informed Consent

Participation is voluntary

It is your choice whether or not to participate in this research. If you choose to participate, you may change your mind and leave the study at any time. Refusal to participate or stopping your participation will involve no penalty or loss of benefits to which you are otherwise entitled.

What is the purpose of this research?

The purpose of this research is to examine human judgments about automated vehicles. All data from this experiment are gathered for scientific purposes and will contribute to our eventual understanding of brain and visual function. These data may be published in scientific journals so that other researchers may have access to these data.

How long will I take part in this research?

Your participation will take approximately 3 minutes to complete.

What can I expect if I take part in this research?

As a participant, you will be asked to look at images presented on a video display and give responses with key presses or movements of a mouse pointing device. Your response may involve responding as quickly as you can, memorizing what you saw, making a judgment, or completing a questionnaire. You will also be asked to complete a demographics form.

What are the risks and possible discomforts?

If you choose to participate, the effects should be comparable to those you would experience from viewing a computer monitor for 15 minutes and using a mouse or keyboard, e.g., eye fatigue. You are free to take breaks throughout the session.

Will I be compensated for participating in this research?

You will be compensated \$0.60 for this study. You will still receive payment if you withdraw early.

If I take part in this research, how will my privacy be protected? What happens to the information you collect?

Your participation in this experiment will remain confidential, and your identity will not be stored with your data.

If I have any questions, concerns or complaints about this research study, who can I talk to?

The researcher for this study is Julian De Freitas who can be reached at 626.559.6401; #161 Morgan Hall, 15 Harvard Way, Boston MA, 02163; jdefreitas@hbs.edu. If you have questions, concerns, or complaints, If you would like to talk to the research team, If you think the research has harmed you, or If you wish to withdraw from the study. This research has been reviewed by the Committee on the Use of Human Subjects in Research at Harvard University. They can be reached at 617-496-2847, 1350 Massachusetts Avenue, Suite 935, Cambridge, MA 02138, or cuhs@harvard.edu for any of the following: If your questions, concerns, or complaints are not being answered by the research team, If you cannot reach the research team, If you want to talk to someone besides the research team, or If you have questions about your rights as a research participant.

☐ I agree (1)

☐ I disagree (2)

End of Block: informed_consent

Start of Block: Prolific ID



prolific_id What is your Prolific ID?

Please note that this response should auto-fill with the correct ID.

End of Block: Prolific ID

Start of Block: attention_checks

att_1 **This is an attention check.**

John is taller than Paul. Who is shorter?

- ☐ John (1)
 - ☐ Paul (2)
 - ☐ Neither John nor Paul (3)
 - ☐ Both John and Paul (4)
-

att_2 **What color is grass? The fresh, uncut grass, not leaves or hay.**

Please make sure to select purple, so that we know you're paying attention.

- ☐ Green (1)
- ☐ Purple (2)

End of Block: attention_checks

Start of Block: attention check fail

Q143 Unfortunately, you have failed to answer at least one of the attention check questions correctly and have been redirected to the end of the survey.
Please click the button to below to be redirected back to Prolific and register your submission.

End of Block: attention check fail

Start of Block: SAE_comps

Q207 Please read the information presented below carefully in order to be able to understand and answer the survey questions

Below is a table depicting 6 possible types of vehicles. The different types are based on how many driving tasks are controlled by the human driver ('HUMAN') versus the self-driving system ('MACHINE').

In level 1, the human controls all driving tasks, whereas in level 6, the machine controls all

driving tasks. The levels in-between differ in which driving tasks are controlled by the human versus the machine.

comp_1 **You were shown a table. What did the table represent?**

- ☐ The number of typical daily tasks controlled by humans vs. machines (1)
- ☐ Different types of vehicles based on how many driving tasks are controlled by humans vs. machines (2)
- ☐ Six people's preferences for humans vs. machines (3)

comp_2 **What were the values for vehicle type #1 (the one at the top of the table)?**

- ☐ MACHINE, HUMAN, HUMAN, MACHINE (1)
- ☐ HUMAN, MACHINE, MACHINE, HUMAN (2)
- ☐ MACHINE, MACHINE, MACHINE, MACHINE (3)
- ☐ HUMAN, HUMAN, HUMAN, HUMAN (4)

End of Block: SAE_comps

Start of Block: (1) autopilot

auto_ad AEON is an automobile firm that is planning on introducing a new automated vehicle program in the market. They have named their program: AEON **AUTOPILOT**.

Please answer some questions so that AEON can understand and analyze preferences for automated vehicles and investigate the potential of their automated vehicle program.

End of Block: (1) autopilot

Start of Block: (1) autopilot_mcq

auto_verb Which of the following do you believe is the best description of the **AUTOPILOT** system when it is switched **on**:

- ☐ Autopilot causes the car to drive (i.e., it fully executes all driving decisions). (1)
- ☐ Autopilot helps the car to drive (i.e., it assists the human with some driving decisions). (2)
- ☐ Autopilot prevents the car from driving (i.e., it counteracts the human's driving decisions) (3)
- ☐ None of the above (4)



auto_explanation Please explain your answer in at least one sentence.

End of Block: (1) autopilot_mcq

Start of Block: (1) autopilot_followup

auto_magnitude In terms of percentages, how much do you think **AUTOPILOT**, as opposed to the human, causes the car to drive when the system is **on**.

Human causes all driving	Autopilot and human equally cause driving	Autopilot causes all driving
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0 10 20 30 40 50 60 70 80 90 100

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End of Block: (1) autopilot_followup

Start of Block: (1) autopilot_capability

perceived_automation What do you perceive as the level of automation for AEON's AUTOPILOT Program? Please refer to the levels of automation table described earlier (see also below).

☐ 1 (1)

☐ 2 (2)

☐ 3 (3)

☐ 4 (4)

☐ 5 (5)

☐ 6 (6)

av_info Click to write the question text

End of Block: (1) autopilot_capability

Start of Block: (1) autopilot_comp3



comp_3 According to the questions, you were asked to rate your impressions of...

☐ Road traffic (1)

☐ Car accidents in your city (2)

☐ Autopilot's driving abilities (3)

End of Block: (1) autopilot_comp3

Start of Block: (2) copilot

co_ad AEON is an automobile firm that is planning on introducing a new automated vehicle program in the market. They have named their program: AEON **COPILOT**.

Please answer some questions so that AEON can understand and analyze preferences for automated vehicles and investigate the potential of their automated vehicle program.

End of Block: (2) copilot

Start of Block: (2) copilot_mcq

co_verb Which of the following do you believe is the best description of the **COPILOT** system when it is switched **on**:

- ☐ Copilot causes the car to drive (i.e., it fully executes all driving decisions). (1)
- ☐ Copilot helps the car to drive (i.e., it assists the human with some driving decisions). (2)
- ☐ Copilot prevents the car from driving (i.e., it counteracts the human's driving decisions) (3)
- ☐ None of the above (4)

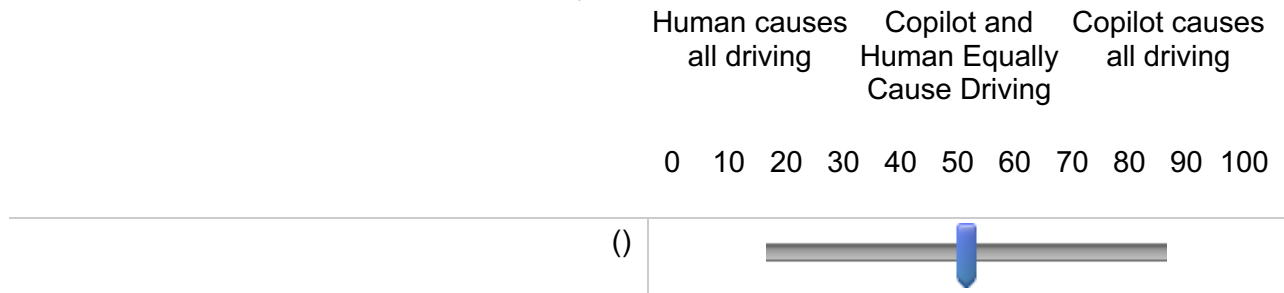


co_explanation Please explain your answer in at least one sentence.

End of Block: (2) copilot_mcq

Start of Block: (2) copilot_followup

co_magnitude In terms of percentages, how much do you think **COPILLOT**, as opposed to the human, causes the car to drive when the system is **on**?



End of Block: (2) copilot_followup

Start of Block: (2) copilot_capability

capability_copilot What do you perceive as the level of automation for AEON's COPILLOT Program? Please refer to the levels of automation table described earlier (see also below).

- ☐ 1 (1)
- ☐ 2 (2)
- ☐ 3 (3)
- ☐ 4 (4)
- ☐ 5 (5)
- ☐ 6 (6)

av_info Click to write the question text

End of Block: (2) copilot_capability

Start of Block: (2) copilot_comp3



comp_3 According to the questions, you were asked to rate your impressions of...

- ☐ Road traffic (1)
- ☐ Car accidents in your city (2)
- ☐ Copilot's driving abilities (3)

End of Block: (2) copilot_comp3

Start of Block: demographics

gender What is your gender?

- ☐ Male (1)
 - ☐ Female (2)
 - ☐ Prefer not to disclose (3)
 - ☐ Other (please specify) (4)
-

ethnicity What is your ethnicity?

- ☐ Black or African American (1)
 - ☐ Asian (2)
 - ☐ White or European (3)
 - ☐ Hispanic or Latino (4)
 - ☐ Mixed (5)
 - ☐ Prefer Not to Say (7)
 - ☐ Other (please specify) (6)
-

age What is your age (in years)?

education Please indicate the highest level of education completed.

- ☐ High School or Equivalent (1)
- ☐ Vocational/Technical School (2 year) (2)
- ☐ Some College (3)
- ☐ College Graduate (4 year) (4)
- ☐ Masters Degree (MS) (5)
- ☐ Doctoral Degree (PhD) (6)
- ☐ Professional Degree (MD, JD, etc.) (7)
- ☐ Other (please specify) (8)

license **Do you have a driving license?**

- ☐ Yes (1)
- ☐ No (2)

ai_knowledge **How much would you say you know about automated cars?**

Very little

A lot

0 10 20 30 40 50 60 70 80 90 100

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End of Block: demographics